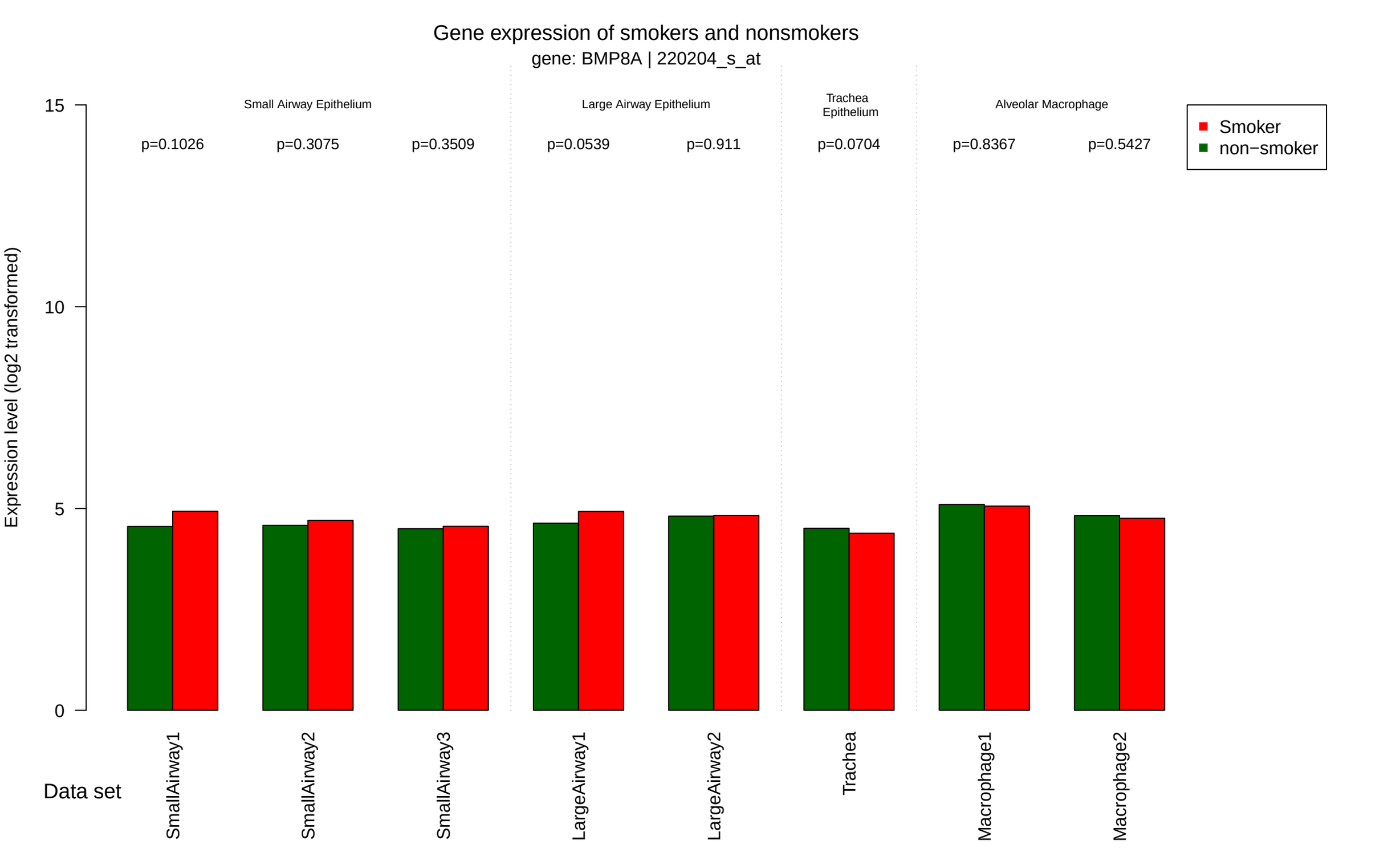
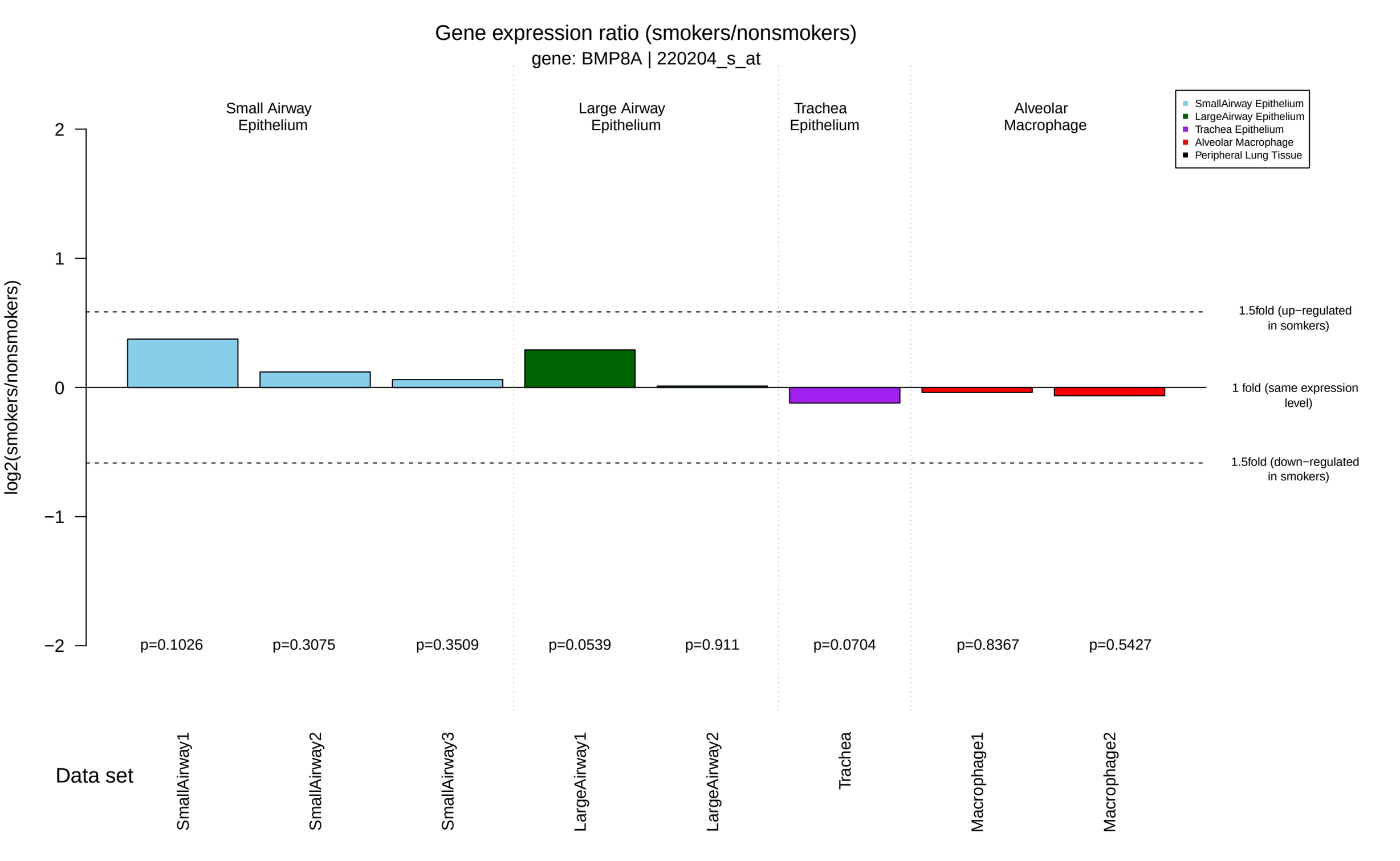
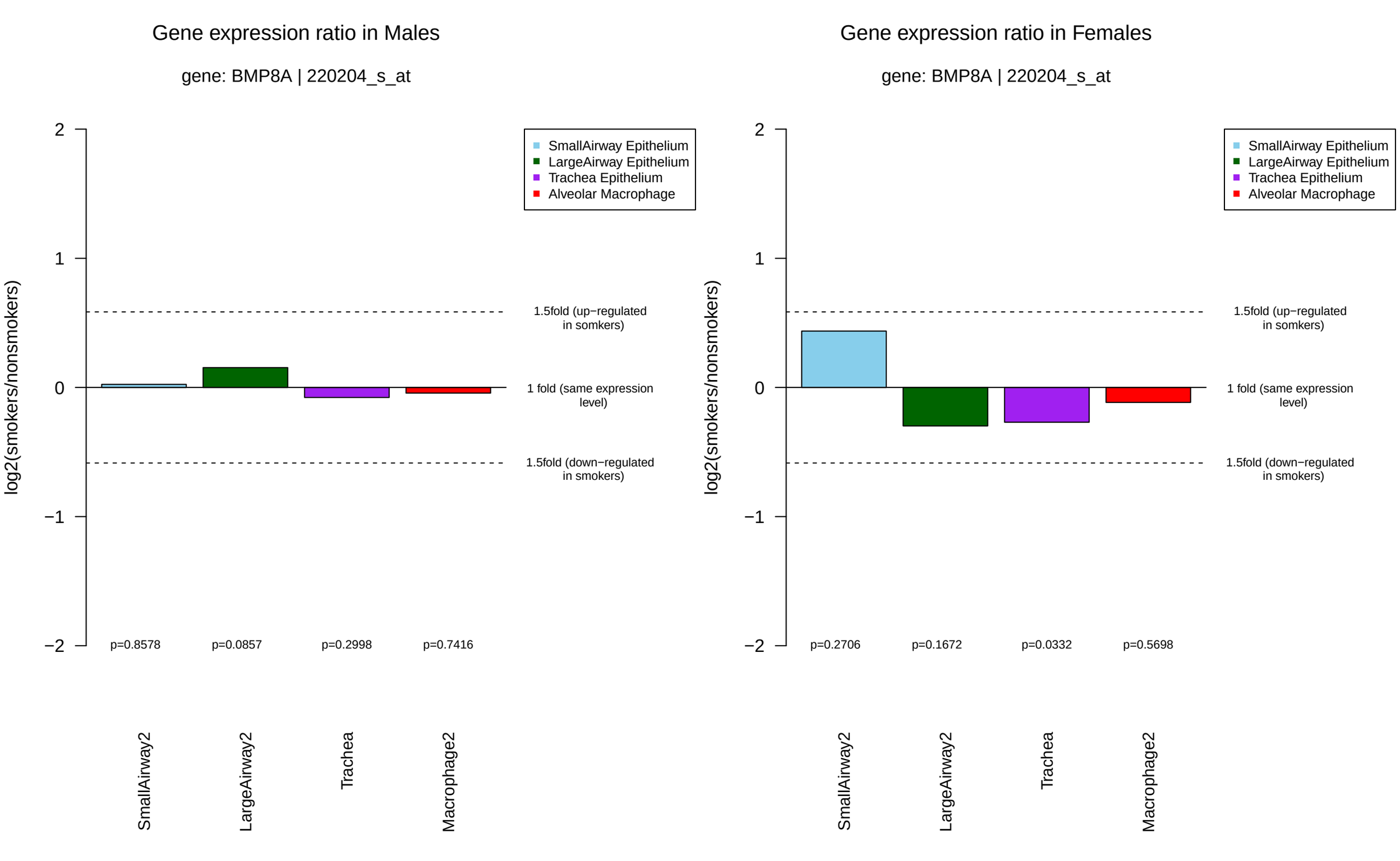
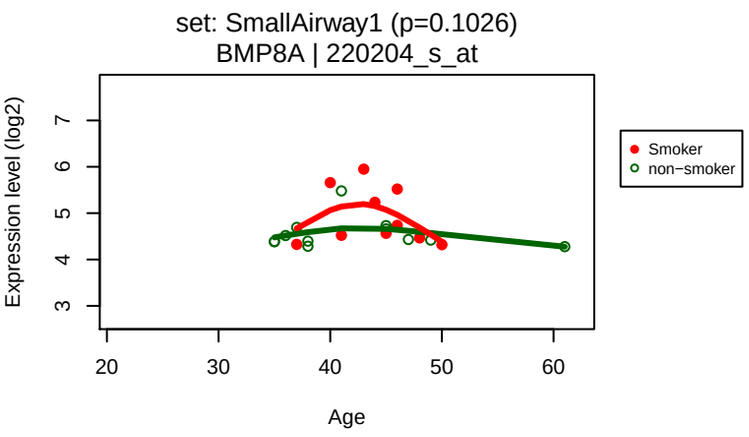
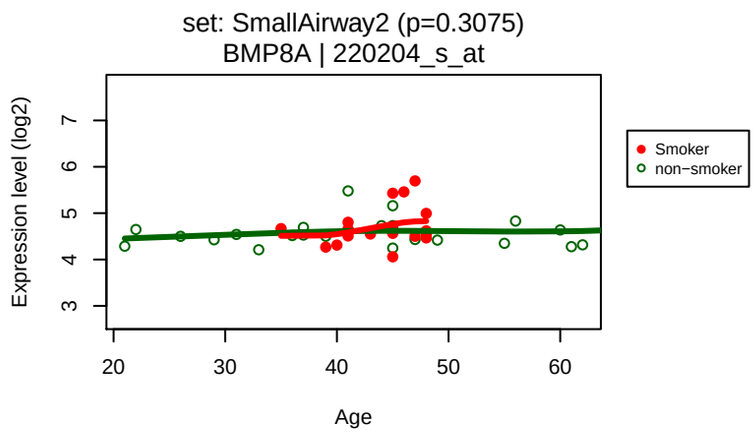
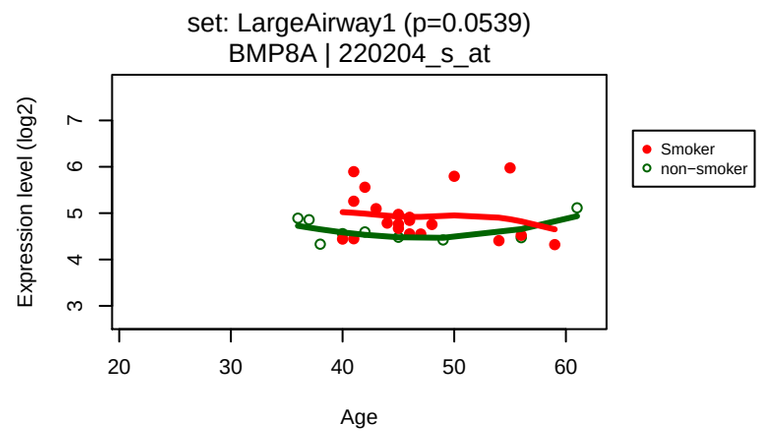

Gene expression of smokers and nonsmokers
gene: BMP8A | 220204_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1026
p=0.3075
p=0.3509
p=0.0539
p=0.911
p=0.0704
p=0.8367
p=0.5427
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

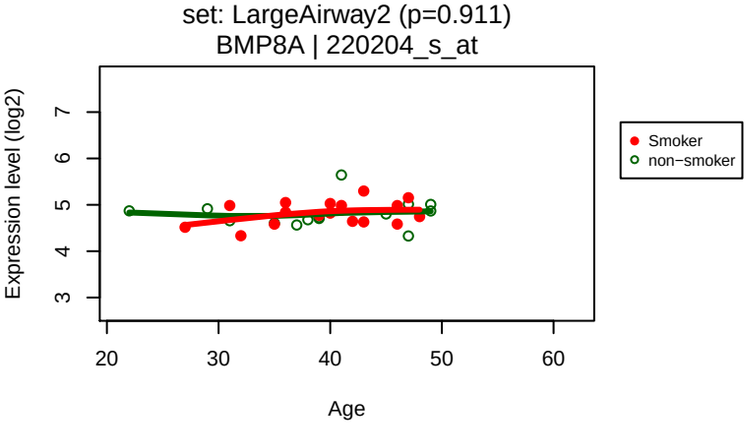

Gene expression ratio (smokers/nonsmokers)
gene: BMP8A | 220204_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1026
p=0.3075
p=0.3509
p=0.0539
p=0.911
p=0.0704
p=0.8367
p=0.5427
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

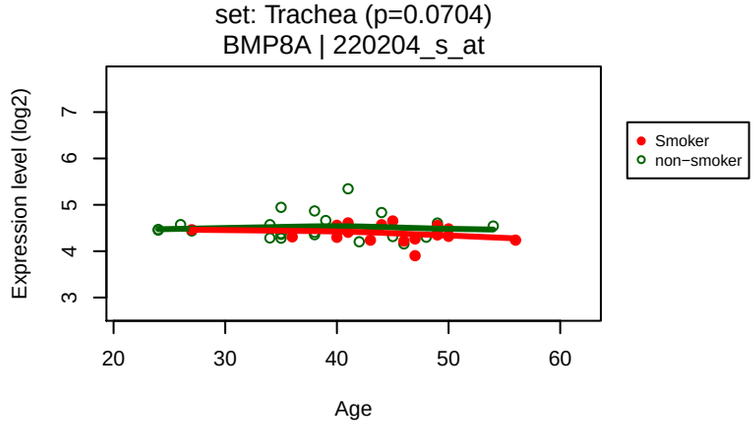

Gene expression ratio in Males
gene: BMP8A | 220204_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8578
p=0.0857
p=0.2998
p=0.7416
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BMP8A | 220204_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2706
p=0.1672
p=0.0332
p=0.5698
SmallAirway2
LargeAirway2
Trachea
Macrophage2

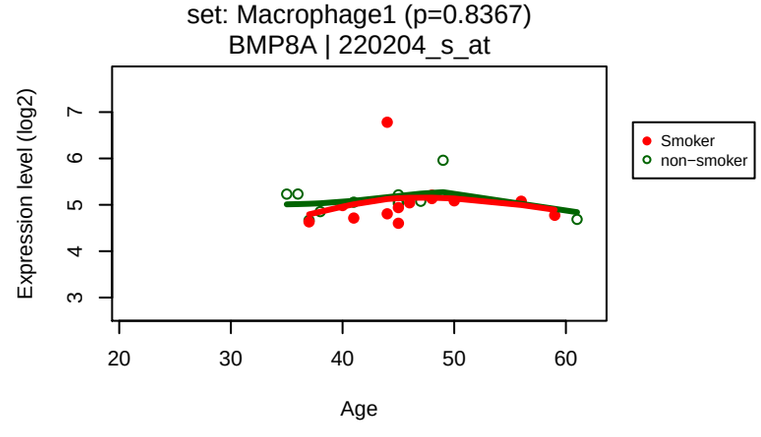

set: SmallAirway1 (p=0.1026)
BMP8A | 220204_s_at
Expression level (log2)
Age
Smoker
non−smoker

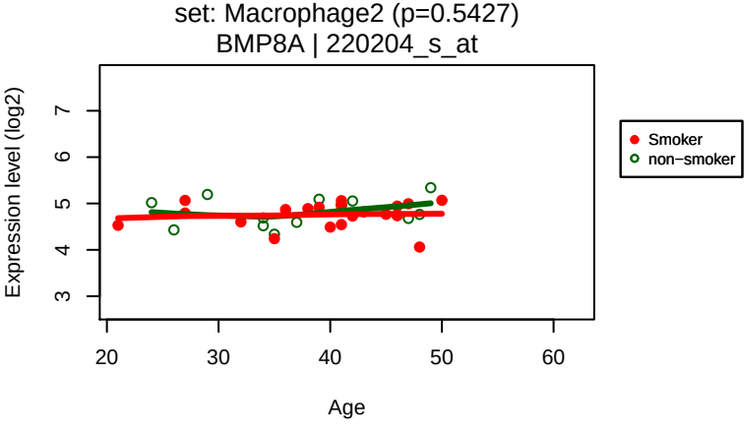

set: SmallAirway2 (p=0.3075)
BMP8A | 220204_s_at
Expression level (log2)
Age
Smoker
non−smoker

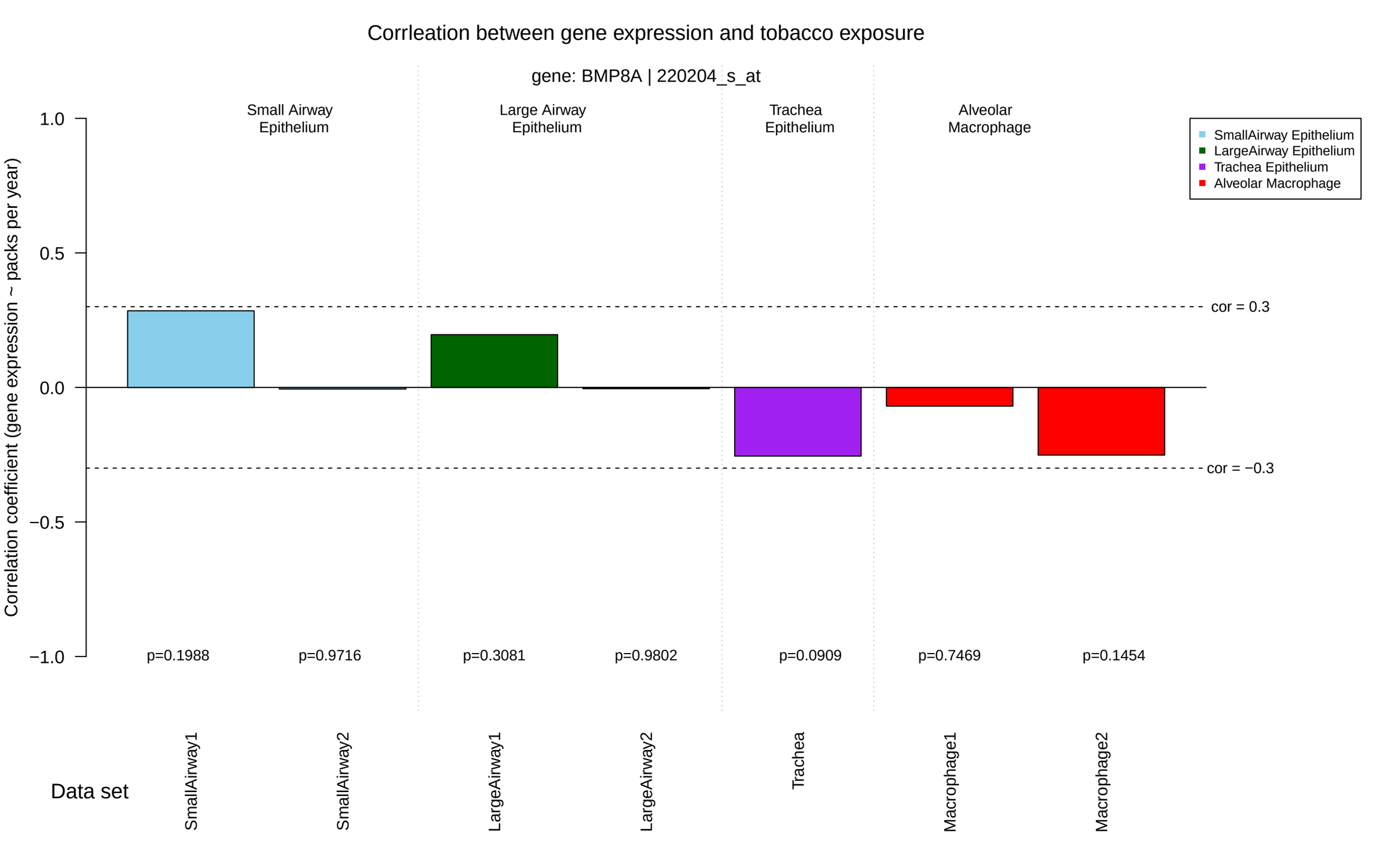

set: LargeAirway1 (p=0.0539)
BMP8A | 220204_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.911)
BMP8A | 220204_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0704)
BMP8A | 220204_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8367)
BMP8A | 220204_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5427)
BMP8A | 220204_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BMP8A | 220204_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1988
p=0.9716
p=0.3081
p=0.9802
p=0.0909
p=0.7469
p=0.1454
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2